Sentence Processing is Surprisingly Robust to Verbatim Repetition

Erxiao Wang[1]*, Ekaterina Ivshina[2], Yuhan Zhang[1] & Cory Shain[1]
[1] Department of Linguistics, Stanford University [2] School of Engineering and Applied Sciences, Harvard University *exw@stanford.edu

Introduction: Verbatim repetition of sentences is almost always avoided in psycholinguistic experiments, and for good reason: decades of research documents many ways that language processing is sensitive to context [1-4], so it stands to reason that an item will be processed qualitatively differently upon repeated presentation. For example, the subjective predictability of words could change dramatically upon re-presentation, potentially altering processing measures [5]. But if processing is in fact similar across repetitions (e.g., is largely automatic and involuntary [6, 7]), then averaging over verbatim repetitions would be a powerful way to improve signal in (and thus, the feasibility of) studies that target detailed item-driven effects, including naturalistic, norming, decoding, and representational similarity studies. Indeed, verbatim item repetition is already widely used to capture stable response properties in vision neuroscience [8], and naturalistic reading suggests that predictability effects are largely conserved under repetition [9]. Motivated by these insights, we directly investigated the effects of sentence repetition in a widely-used psycholinguistic paradigm (word-by-word self-paced sentence reading, *SPR*) as well as the potential for repetition to improve signal quality.

Methods: We ran a large-scale crowd-sourced SPR experiment (n=988). Five repetitions of 40 unique sentences (items) were randomly interleaved with 100 filler items, for a total of 300 exposures per participant. Sentences were drawn at random from the set of 12-word "baseline" sentences of ref. [10]. Because we were primarily interested in the effect of repetition on item-level responses, no other experimental manipulation (beyond repetition) was applied.

Effects of repetition on reading: We modeled by-word response time (RT) across the entire item as a function of word length, frequency, GPT-2 surprisal, and repetition index, using continuous-time deconvolutional regressive neural network (CD-RNN) models, which estimate nonlinear effects and interactions over time [11]. Across repetitions, the qualitative pattern of psycholinguistic effects was preserved: long, low-frequency, and high-surprisal words had reliably longer RT on every exposure. However, all effects showed subtle but reliable attenuation with repetition, especially for low-frequency words. We found statistically significant interactions between repetition and each linguistic predictor (1–5) (Figure 1).

Effects of repetition on signal quality: We then studied the extent to which the repetition-invariant component of the response can be used to improve signal in SPR experiments. To do so, we first regressed out by-participant intercepts and general nonlinear trends [12]. We then measured signal quality between and within participants. Between participants, we found that a given participant's response to an item at initial presentation was more strongly correlated to the other participants' average response across repetitions of that item than it was to the other participants' initial responses to that item (Figure 2). Correspondingly, item decoders are more accurate when trained on the averaged responses across repetitions than when trained on initial responses (Figure 3). Within participants, we found using leave-one-item-out cross-validation (LOOCV) that a linear model using the three linguistic predictors above better predicted the initial response to an item when the model had access to repeated items during fitting (Figure 4). Finally, we found that linear models of averaged repetitions showed generally stronger psycholinguistic effects than linear models of initial presentations, even though both settings were equal in data quantity (Figure 5).

Conclusion: Our results suggest that word-by-word SPR in simple sentences is surprisingly robust to verbatim repetition of items. Although linguistic effects reliably attenuate, the attenuation is subtle, and the qualitative pattern of processing is largely conserved. Moreover, repetition improves diverse measures of item-level signal quality, suggesting that repetition can improve the viability of detailed item-level behavioral studies (e.g., decoding) in psycholinguistics.

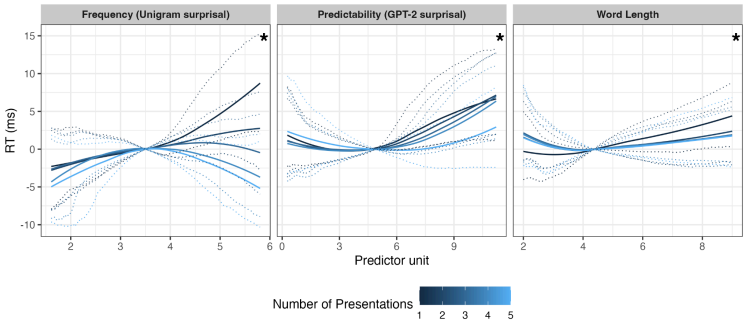

Figure 1. CDRNN estimated effects of frequency (Unigram surprisal), predictability (GPT-2 surprisal) and Word Length on word-by-word reading time across repetitions. Plots show how the response at the delay of 0.35s (i.e., 0.35s after the word appear on screen) deviates from its mean as a function of one predictor at the first and repeated presentations (1–5). Dotted lines show 95% variational Bayesian credible intervals. Plots with significant effects-repetition interaction are marked with *. For each predictor, effects on RT are similar across repetitions and reduce by magnitude, supporting an attenuative effect.

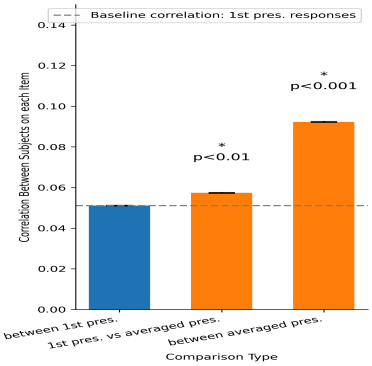

Figure 2: Model free between-subject response correlation on items. Baseline is set to reflect correlation between first presentation responses on each item. Results show significant differences between 1) baseline and correlation between 1st presentation and averaged responses; 2) baseline and correlation scores within averaged responses.

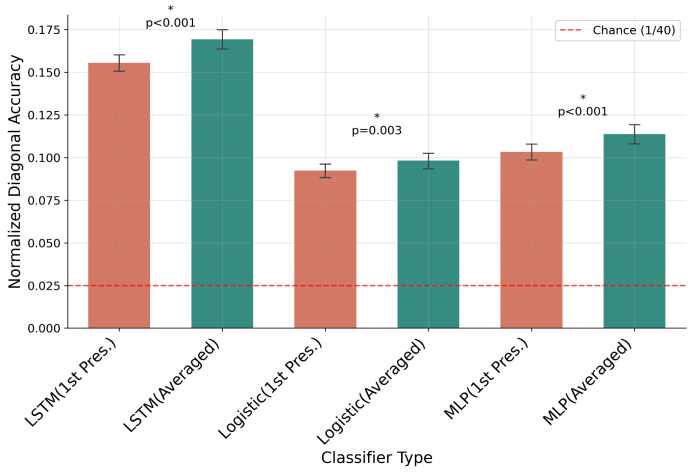

Figure 3: Comparison of per-item decoding accuracy between first presentation and averaged presentations across three types of classifier: Logistic Regression, MLP, and LSTM. * indicates statistically significant differences ($p < 0.05$) from paired t-tests.

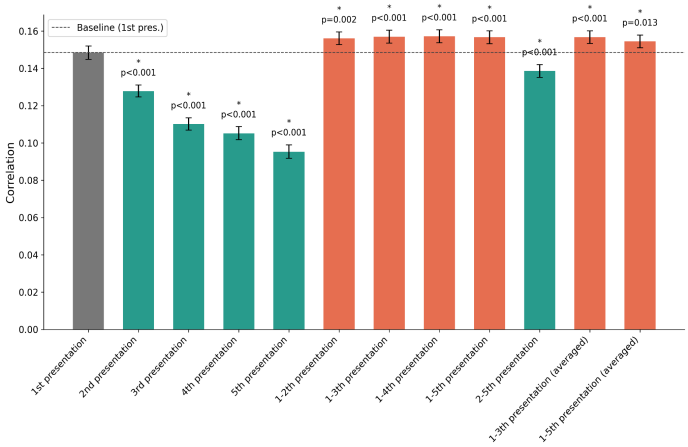

Figure 4: Correlation between LOOCV inference from different subsets of response and the actual response (1st presentation). * indicate statistically significant differences from the baseline (1st presentation).

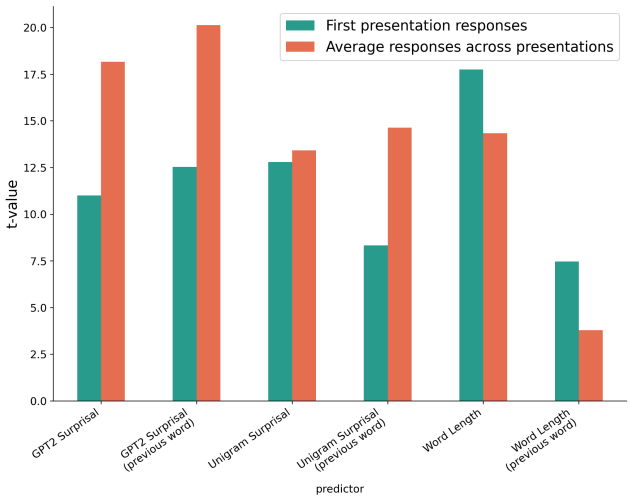

Figure 5: Comparison between Linear Mix Effect Models (LMEM) t-statistics for predictors.LMMs were trained on 1st presentation responses and averaged responses, respectively.

References:
[1] Kutas, M., & Hillyard, S. A. (1980). *Science*. [2] Tabossi, P., Spivey-Knowlton, M. J., McRae, K. *et al.* (1994). *Attention and Performance.* [3] Smith, N. J., & Levy, R. (2013). *Cognition*. [4] Lowder, M. W., Choi, W., & Gordon, P. C. (2013). *Memory & Cognition*. [5] Staub, A. (2015). *Language and Linguistics Compass*. [6] Pulvermüller, F., & Shtyrov, Y. (2006). *Progress in Neurobiology*. [7] Shtyrov, Y., MacGregor, L. (2016). *Sci Rep*. [8] Isik, L., Meyers, E. M., Leibo, J. Z. *et al.* (2014). *Journal of Neurophysiology*. [9]] Klein, K. G., Frenkel, S., Shubi, O. *et al.* (2025). *In arXiv [cs.CL]. arXiv.* [10] Tuckute, G., Sathe, A., Srikant, S. *et al.* (2024). *Nat Hum Behav*. [11] Shain & Schuler (2024). *Open Mind*. [12] Prasad & Linzen (2021). *JEP: LMC*.